WILSON

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Banking Technology Leader — Payment Middleware & Productised SaaS | Solution Design to Delivery

PROFESSIONAL SUMMARY

Banking technology leader who designs and delivers payment middleware between a Tier-1 bank's channels and its core banking — validation, enrichment and ISO 20022 messaging (SWIFT's MT-to-MX migration) — and has taken a configurable SaaS product from 0 to 1, with 18+ years across software engineering, architecture and delivery. Vice President at UOB since 2022, leading 25 onshore and offshore engineers on payment and lending platforms serving millions of accounts.

Began as an engineer and technical lead on e-commerce and news-content platforms at Renren, Sohu and Baidu, then spent 2016–2022 in head-of-technology and architect roles at start-ups and scale-ups — heading technology at a smart-parking IoT company, co-founding an audio-learning platform and leading VV Technology's 140-person technology organisation through its managers.

On the commercial side, has led winning technical bids as a vendor and, at UOB, holds decision or joint-decision authority on vendor selection as the buyer. Delivers end to end — requirements, solution design, testing, release and production support — under banking-grade change, release and access controls.

CORE COMPETENCIES

Banking Solutions	Payment initiation, validation, enrichment and status tracking; ISO 20022 (MT-to-MX) messaging migration; core-banking integration; retail lending servicing and collections
Solution Architecture	Payment middleware, API design, microservices, Apache Camel, event-driven integration, service orchestration, AI workflow automation, legacy modernisation with phased cut-over
Productisation	Configurable multi-tenant SaaS from 0 to 1, reusable backend services, a SaaS model for business and public-sector customers, release planning and governance
Bids & Commercial	Technical bids and tender responses; buyer-side vendor evaluation and selection, SOW/licence negotiation, vendor performance reviews; delivery and technology budgets
Governance & Quality	Secure SDLC, role-based access control, change control, release governance, SIT/UAT sign-off, disaster recovery (DR) and failover, MAS TRM (Singapore technology-risk rules)
End-to-End Delivery	Requirements to production support; Agile/Scrum and hybrid Waterfall; risks, dependencies and scope trade-offs; data migration; steering-committee reporting
Technical	Java, Spring Boot, Spring Cloud, Apache Camel, Kafka, Kubernetes, Docker, AWS, Linux (RHEL), SQL

SELECTED ACHIEVEMENTS

- **Payment middleware in a Tier-1 bank:** Led end-to-end design of a payment-initiation hub — API contracts, validation, enrichment and status tracking — that connects channels and vendor systems and hands payments to core banking for settlement.
- **Modernisation without disruption:** Planned and architected the phased move of cross-border payments to ISO 20022 (SWIFT's MT-to-MX migration), running old and new message formats side by side: new-format messaging went live in May and August 2025, ahead of SWIFT's November 2025 deadline, and the old path was retired in August 2026 with no service interruption.
- **Winning technical bids:** At AI Park, led technical bids that won city parking projects in Shijiazhuang, Handan and Beijing, and launched pilots in Shijiazhuang (2016), Handan (2017) and Beijing Chaoyang (2017).
- **Buyer-side commercial authority:** At UOB, hold decision or joint-decision authority on technology investment, vendor selection and SOW/licence negotiation, and chair budget and performance reviews for four IT service vendors.
- **Configurable SaaS product:** Architected and delivered, from 0 to 1, a multi-tenant SaaS platform for online-game businesses — reusable services for KYC, bonuses, VIP tiers, analytics and payments — on Spring Cloud, Kubernetes and AWS.
- **Banking-grade governance:** Established release governance with SIT/UAT sign-off for every release, designed role-based permissions and encryption under MAS TRM, and personally led DR and failover drills for the real-time payment middleware, sustaining 99.9%+ availability.

PROFESSIONAL EXPERIENCE

United Overseas Bank (UOB) | Singapore — Oct 2022 – Present

Vice President (VP2) / Technical Delivery Manager — Real-Time & Payment Data Services, Group Technology & Operations

Technology leader for two mission-critical platform lines in a MAS-regulated Tier-1 bank, serving millions of accounts: cross-border payments (SWIFT ISO 20022 messaging) and retail unsecured lending — payment initiation, validation and lifecycle tracking, plus post-disbursement collections across credit-card and unsecured-loan portfolios.

Solution Architecture & Engineering

- Personally built the payment-initiation hub's core modules — payment-transaction initiation, concurrency control, the corporate- and user-role permission matrix and processing of payment status results returned by downstream systems.
- Reduced end-to-end transaction latency under high load by tuning thread pools and the Java virtual machine (JVM) and removing synchronous blocking calls.
- Lead the 10-engineer backend team of a regional customer-360 platform (in production since 2022) that takes daily account summaries from core banking for portfolio analysis; personally built its data-analysis module and Spring Batch jobs.
- Designed and built two AI workflow automations: a code-review service — live since May 2026 after formal architecture and risk approval, with humans keeping merge authority — and an agent that removed a manual document-reconciliation task.

Delivery, Governance & Support

- Own end-to-end delivery of the ISO 20022 programme with a 15-engineer onshore/offshore team — scope, roadmap, risks, dependencies on channels and vendor systems, release planning and production rollout — shipping upgrades every 2–3 months.
- Led data migration for the payment-initiation hub, the retail-lending platform and the customer-360 platform, with sensitive-data encryption and credential management.
- Enforce CI/CD quality gates — JUnit, a 99% SonarQube coverage gate on all code, automated regression and penetration testing — and run secure SDLC, data protection impact assessments (DPIA) and data governance in line with MAS TRM.
- Coordinate P1–P3 incidents end to end — escalation, stakeholder communication, root-cause analysis (RCA) and post-incident reports; with no separate SRE team, the platform teams own production support, service-level indicators/objectives and Splunk monitoring.

Commercial & Stakeholders

- Lead technical evaluation of vendors as input to vendor-selection and SOW/licence decisions.
- Resolve schedule conflicts between the two platform lines by agreeing MVP scope with the requirement, project and technical leads, confirming dependencies and setting a sequence all parties accept.
- Act as the primary interface between business stakeholders, product owners, operations, compliance, architects and engineering managers; present status, risks and decisions to senior management and the steering committee.

Golden Bridge Drive (SG) Pte. Ltd. | Singapore — Nov 2020 – Jul 2022

Project Manager & Architect

- Led a 40-person project team, including outsourced staff, introduced Agile/Scrum from scratch and CI on Jenkins and Amazon ECR, and owned delivery budget, capacity and launch readiness for the multi-tenant SaaS platform.
- Ran hybrid delivery — Waterfall for scope, budget, milestones and resource planning; Agile sprints for execution, backlog tracking and iterative delivery.
- Designed hierarchical KYC identity verification and configurable rule models for authentication levels and for bonus eligibility, triggers and distribution.

VV Technology Pte. Ltd. | Singapore — Apr 2019 – Jun 2020

Head of Technical Centre

VV-Life: a local-life services platform connecting offline merchants — food, fresh produce, travel and everyday services.

- Led VV's 140-person technology organisation through its managers and built its microservices platform from scratch, owning technology selection and the technology budget.
- Delivered the first VV-Life release in 2019 and architected its payment subsystem, integrating UnionPay, Visa/Mastercard, Google Pay and GrabPay payments with settlement reconciliation.
- Designed the shared back-end platform architecture (Spring Boot, MySQL cluster, Redis, RocketMQ) for a B2B online-office product serving Southeast Asia, governing solution design, cost evaluation, staffing and acceptance.
- Introduced CI/CD, code standards and containerised deployment with rollback on Kubernetes; defined production-incident priority levels and tiered emergency-response plans.

Hengshunyuan Information Technology Co., Ltd. | China — Oct 2017 – Sep 2019

Co-founder

- Owned product, architecture and technical delivery of an audio-learning platform, reaching a peak of ~100k daily active users; ran content licensing with rights holders, bringing ~1,500 licensed titles and albums onto the platform at launch.

AI Park | China **Apr 2016 – Sep 2017**

VP of Technical Department

- Led architecture, delivery, budget and the technical team for a smart-parking IoT platform connecting car parks to the cloud for automated management, online payment and recommendations.
- Introduced a SaaS model for future business and government customers.
- Brought Spring Cloud microservices into the architecture, with Kafka for asynchronous flows and caching to shield the database, achieving millisecond-level service processing.

EARLIER EXPERIENCE

- **Baidu — Technical Specialist, Nuomi group-buying e-commerce platform (May 2014 – Dec 2014, China):** As project leader, re-architected the merchant-side order system and delivered two months ahead of plan; designed and coded Nuomi's payment-gateway integration with Alipay, WeChat Pay and UnionPay.
- **Sohu.com — Technical Leader / Project Leader, Sohu News content platform (Mar 2012 – May 2014, China):** Led the re-architecture of the fifth-generation CMS onto Spring MVC and a Redis cluster, and delivered its OpenAPI messaging platform.
- **Renren Network — Senior R&D Engineer (Jul 2008 – Mar 2012, China):** Developed Renren's shopping website, Renren Mall and CRM, and designed and coded Renren Mall's payment integration with Alipay and UnionPay.

EDUCATION, CERTIFICATIONS & LANGUAGES

- Master's degree in Computer Science and Technology — Liaoning University (2005 – 2008)
- Bachelor's degree in Applied Mathematics — Shenyang University of Technology (2000 – 2004)
- Software Designer (Intermediate) — Qualification Certificate of Computer and Software Technology Proficiency (2006) · PMP — in progress · Site Reliability Engineering (SRE) — formal training
- Languages: English and Mandarin (professional working proficiency in both)